

# High-Value Roles, Skills, and Target Companies for ECE & EEE Students

## Embedded / Firmware Roles

Roles: Embedded Systems Engineer, Firmware Engineer, BSP Engineer, Device Driver Engineer

Skills: Embedded C, Microcontrollers, RTOS, UART/SPI/I2C, Debugging

Companies: Texas Instruments, STMicroelectronics, NXP Semiconductors, Microchip Technology

## VLSI / Chip Design Roles

Roles: VLSI Design Engineer, Physical Design Engineer, RTL Design Engineer, Verification Engineer, DFT Engineer

Skills: Verilog/SystemVerilog, Digital Electronics, STA, Timing Analysis

Companies: Intel, Qualcomm, AMD, NVIDIA

## Automotive / ECU Roles

Roles: ECU Software Engineer, Automotive Embedded Engineer, AUTOSAR Engineer, CAN/LIN Network Engineer

Skills: Embedded C, CAN Protocol, AUTOSAR Basics, Diagnostics

Companies: Bosch, Continental, Valeo, Tata Elxsi

## Power Electronics / EV Roles

Roles: Power Electronics Engineer, EV Systems Engineer, BMS Engineer, Motor Control Engineer

Skills: Power Electronics, Control Systems, MATLAB/Simulink, Inverters, Converters

Companies: Siemens, ABB, Schneider Electric, Ather Energy

## Industrial Automation / PLC Roles

Roles: PLC Programmer, SCADA Engineer, Industrial Automation Engineer, Control Systems Engineer

Skills: PLC, SCADA, Ladder Logic, Sensors & Actuators

Companies: Siemens, Rockwell Automation, Mitsubishi Electric

## Networking / Telecom Roles

Roles: Network Engineer, RF Engineer, Telecom Engineer, 5G Testing Engineer

Skills: Communication Systems, Networking Basics, RF Concepts

Companies: Ericsson, Nokia, Cisco

## **Hardware Testing / Validation Roles**

Roles: Hardware Test Engineer, Validation Engineer, PCB Test Engineer

Skills: Oscilloscope, PCB Debugging, Signal Analysis, Testing Methodologies

Companies: Intel, AMD, Flex